Debaditya Hait

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AI / Machine Learning Engineer | Agentic Systems, AI Security & Computer Vision

Summary

Computer Science and Data Science undergraduate developing applied ML, agentic workflows and AI safety systems. At Samsung, fine-tuned and benchmarked a ModernBERT prompt-injection detector; at IIT Madras, developed physics-informed neural networks. Projects span risk/pricing orchestration, agent permissions, GitHub automation and mobile medical imaging.

Education

SRM Institute of Science and Technology 2023-Present
B.Tech., Computer Science and Engineering – CGPA: 8.5/10 *Kattankulathur, India*

Indian Institute of Technology Madras 2024-Present
B.S., Data Science and Applications – Online degree program *Chennai, India*

Technical Skills

Programming & ML: Python, C/C++, Go, TypeScript, SQL, PyTorch, TensorFlow/TFLite, Hugging Face

Modeling & evaluation: ModernBERT, EfficientNet, computer vision, segmentation, PINNs, fine-tuning, quantization, adversarial evaluation

Agentic AI & security: LangGraph, LiteLLM, tool orchestration, CEL policies, information-flow checks, prompt-injection detection

Backend & data: FastAPI, Pydantic, SQLAlchemy, SQLite, React/Vite, Docker, REST/streaming APIs

Quality & operations: precision/recall/F1, regression testing, GitHub Actions, approval-gated operations, audit logging

Experience

Samsung R&D Institute India Apr 2025-Sep 2025
PRISM Research Intern — LLM Security *Remote*

- Fine-tuned a 395M-parameter ModernBERT encoder for LLM/agent prompt-injection and jailbreak detection using 75,000+ augmented adversarial and benign prompts.
- Evaluated against 315 adversarial prompts: 92.70% accuracy, 0.9021 F1 and 35 ms inference; about 300x faster than generation-based alternatives in the recorded comparison.
- Compared the encoder approach with NeMo Guardrails, Llama Guard 4, Prompt Guard and DeBERTa-v3 under the specified benchmark conditions.

Indian Institute of Technology Madras Jun 2025-Aug 2025
AI Research & Prototyping Intern *On-site, Chennai*

- Developed physics-informed neural networks in PyTorch to model Alzheimer's progression from longitudinal MRI features and biological differential-equation constraints.
- Applied automatic differentiation, collocation sampling and adaptive loss weighting; reported 27% higher early-prediction accuracy versus research deep-learning baselines.

Selected Projects

Quantheon | *Local agentic derivatives pricing and risk desk* | GitHub
Python / FastAPI / LangGraph / SQLAlchemy / React/Vite

- Built a local derivatives desk to price structured products, inspect portfolio valuations/Greeks and scenarios, and review hedge proposals and backtests.
- Orchestrated pricing, risk and research agents through LangGraph and FastAPI, integrating SQLAlchemy persistence with the bundled third-party-credited QuantArk 0.3.0 pricing engine.
- Implemented RFQs, confirmation review, lifecycle events and settlement-cashflow workflows with approval-gated write actions and audit history; sample trading data are illustrative.

BoundaryLab | *Local AI-agent tool-permission and investigation system* | GitHub
Go / CEL / SQLite / Unix sockets / React / WebSockets

- Built a local policy system for hooked AI-agent tool calls that returns allow, deny or approval decisions and exposes rule matches, sessions and security events for investigation.
- Implemented Go hook/daemon enforcement with CEL policies, information-flow and secret/delegation checks, atomic reloads and SQLite audit logs; hooks deny when enforcement is unavailable.
- Tested a 784-case adversarial corpus (278 held out); category recall varied from 93.6% for direct attacks to 38.1% for protocol attacks.

MaintainerOS | *Self-hosted GitHub issue and PR triage assistant* | GitHub
Python / FastAPI / LangGraph / LiteLLM / SQLite / Docker

- Developed a maintainer assistant that prioritizes GitHub issues/PRs, identifies potential duplicates and quality signals, analyzes CI failures, drafts replies and produces digests.
- Combined rule-based triage with optional LangGraph/LiteLLM workflows, TF-IDF matching, SQLite memory and network-isolated Docker Python reproducers; GitHub writes require explicit approval.

WoundDoc | *Mobile medical-imaging ML research* | GitHub
EfficientNet / Segmentation / TFLite / React Native / Expo

- Developed a mobile workflow for wound-type classification, segmentation, calibrated area measurement and longitudinal follow-up in low-connectivity clinical settings.
- Quantized an EfficientNet classifier from 48 MB to 12.5 MB for offline TensorFlow Lite inference; evaluated the mobile workflow on 312 clinician-annotated images from 18 patients at two rural clinics (research pilot).

Open-Source Engineering

Selected upstream ML-framework, embedding and tooling contributions; merge status below reflects the recorded October 2026 snapshot.

TensorFlow (PR #124961) | *Merged (Oct 2026 source record)* | *C++ / oneDNN*

- Prevented oneDNN convolution layout passes from crashing on out-of-range 64-bit attributes by returning validation errors.

Haystack Integrations (PR #3543) | *Merged (Oct 2026 source record)* | *Python / embeddings*

- Fixed int8/uint8 sentence-transformer quantization for single-text requests that could otherwise produce zero vectors and invalid downstream computations.

Docling (PR #3875) | *Merged (Oct 2026 source record)* | *Python / document AI*

- Guarded image enrichment for pageless/plain-text documents, preventing missing-page metadata exceptions; added regression tests.

Astral Ruff (PR #28186) | *Merged (Oct 2026 source record)* | *Rust / Python tooling*

- Corrected false-positive RUF061 diagnostics for pytest assertion helpers used as context managers.

Competitive Programming

- Codeforces: Expert (current); previously held Candidate Master at peak before returning to Expert.
- LeetCode: Guardian; top 1%.
- CodeChef: 5-star.

Publications, Achievements & Certifications

- **Publication:** AI Guardrails: Performance-Optimized Guardrail System for Secure LLM and Agentic AI Interactions — accepted to ICSCCC 2026 (source record).
- **Publication:** WoundDoc: A Mobile AI-Assisted Wound Assessment System for Segmentation, Type Detection, and Longitudinal Healing Monitoring — submitted to MIDL 2026; not reported accepted.
- **Achievement:** Amazon ML Summer School 2026: selected among the top 3,000 of 134,000+ applicants.
- **Achievement:** Goldman Sachs India Hackathon 2026 (Quant): all-India rank 42 of approximately 14,000.
- **Certification:** Microsoft Certified: Machine Learning Operations Engineer Associate (earned Oct 2026).
- **Certification:** Microsoft Certified: Azure AI Apps and Agents Developer Associate (earned Oct 2026).
- **Certification:** Microsoft Certified: Azure AI Engineer Associate (earned Jun 2025).
- **Certification:** GitHub Certified: Agentic AI Developer (GitHub-issued; earned Oct 2026).
- **Certification:** Microsoft Certified: Azure Data Scientist Associate (earned Jun 2025).